The Demand and Considerations for Distributed Inference in the Healthcare Industry

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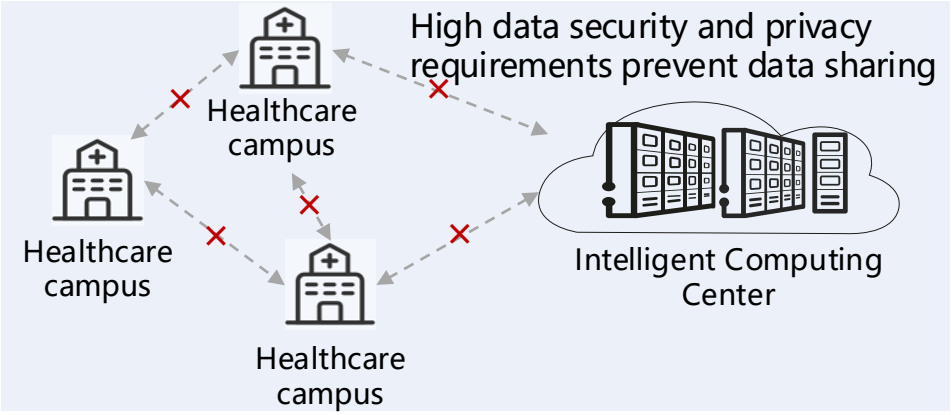
Two Pain Points in the High-Quality Development of the Healthcare Industry in the AI Era

Pain point in data usage

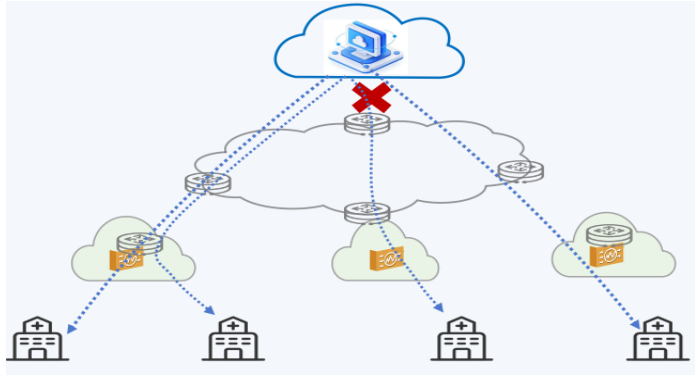
Due to high data privacy and security requirements, secure data sharing cannot be achieved between medical institutions and between medical institutions and computing centers.



Data privacy: The challenge of coordinating the centralized deployment of computing power at intelligent computing centers with the privacy and security of medical data.

Pain point in model usage

High-quality healthcare foundation models deployed in a centralized manner cannot be efficiently and securely applied at all levels of medical institutions across the country due to the lack of computing and network resources.



Model deployment: Dispersed healthcare institutions face challenges in efficiently and securely providing large model inference services.

Existing Models and Challenges of Secure Computing

Existing model 1:
On-premises computing power

Computing power servers are deployed in an on-premises equipment room, which must have sufficient resources such as power, space, and heat dissipation.

Challenge: Fixed and inflexible computing power

1. On-premises all-in-one appliances support only 8 or 16 cards, while enterprises' computing needs are increasing, and the local computing power cannot be elastically expanded.
2. The power supply and heat dissipation of the enterprise equipment room cannot meet the needs of deploying incremental servers.

Existing model 2:
Leasing computing power from carriers

Computing power on the cloud of a carrier can be leased to meet elastic scalability requirements, but data must be sent out of the campus and stored on the cloud for computing.

Challenge: Unsafe data transfer out of the domain

1. Data transfer out of the campus is complex due to the approval process;
2. Data may be leaked when being transferred out of the campus.

Application scenarios of large models in the healthcare industry

Electronic medical record Generation Intelligent triage

Rare diseases/precision medicine Gene sequencing

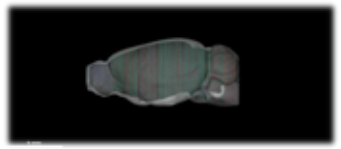
Research Assistant

Clinical Assistant

Imaging diagnosis

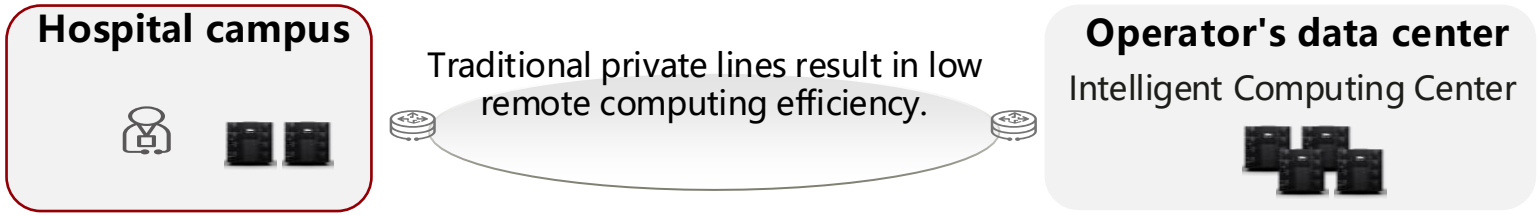
Pathological diagnosis

Multimodal diseases Auxiliary diagnosis



Spatial-temporal transcriptomics

Demands and Considerations for Distributed Inference in the Healthcare Industry

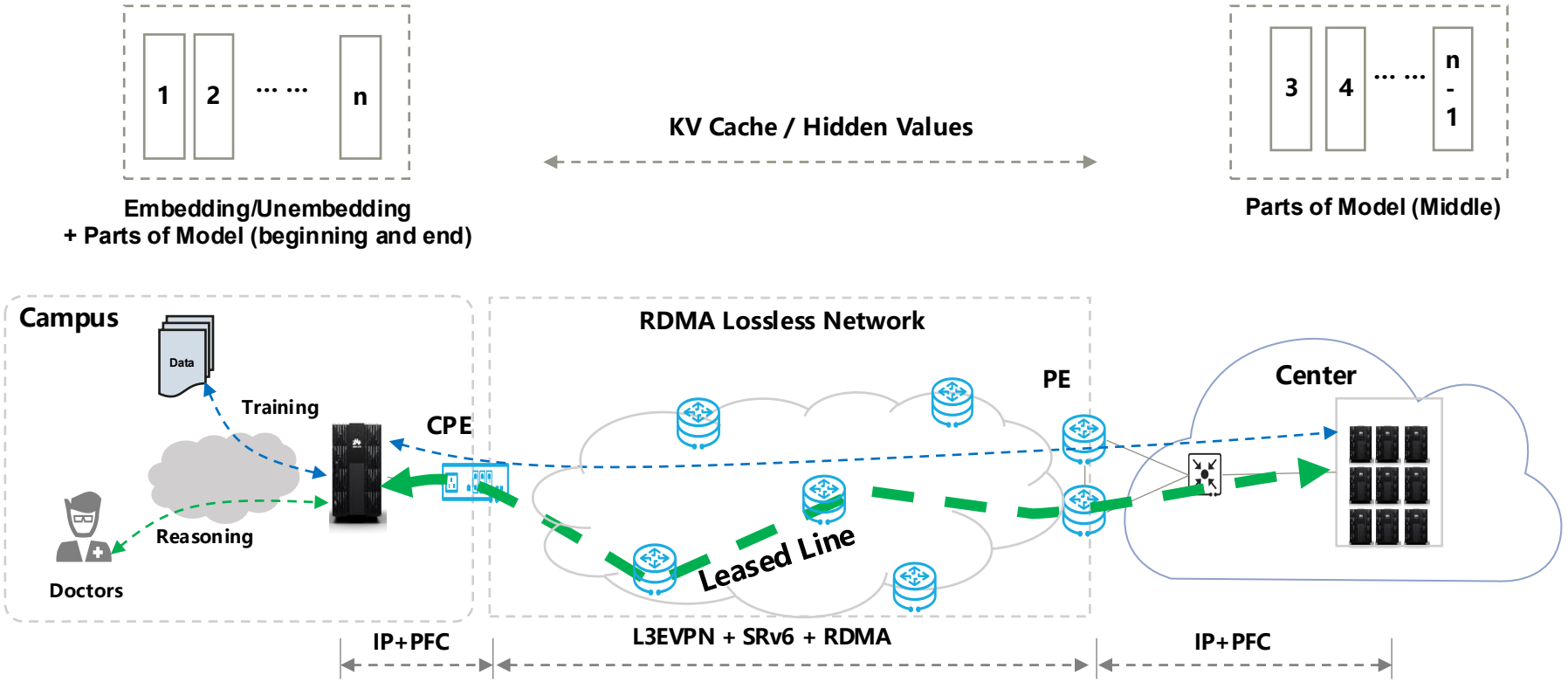


Two self-built servers are not enough for computing power.

There are security concerns when using computing power from other sources.

- **Requirement 1:** Original data must not leave the domain to ensure the security of medical data and prevent data leakage.
- **Requirement 2:** The local computing resources are limited, so elastic computing resources on the cloud can be rented to meet the requirements of medical foundation model applications.
- **Requirement 3:** The computing efficiency of collaborative inference between on-premises and cloud computing resources should be greater than 95%.

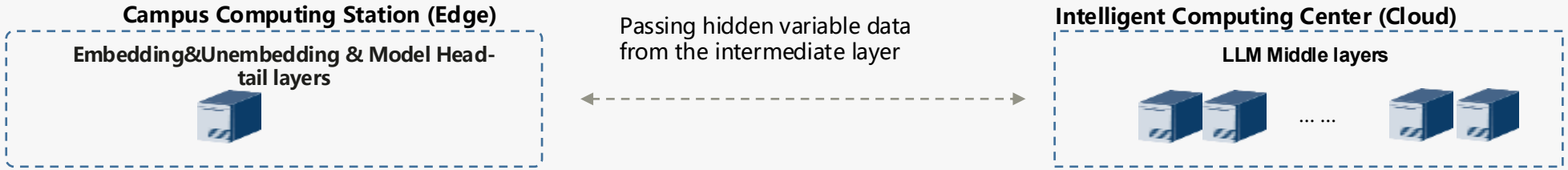
Distributed Training-Reasoning WAN Architecture and Deployment Scenarios



- Through edge-cloud collaborative training and reasoning, prompts are input locally and tokens are generated locally, ensuring data security and preventing it from leaving the domain.
- Hidden variable data is transmitted intermediately, and high-dimensional vectors transformed by the model ensure that the transmitted data is available but invisible.
- Deploying the model's intermediate layers in the cloud prevents the complete model from being restored, ensuring cloud-based model security.
- The larger the model parameter scale, the longer the input/output sequences, and the more layers deployed on the client side, the stronger the data security defense capabilities.

LLM layered deployment, tokens generated locally, prompts do not leave the hospital area

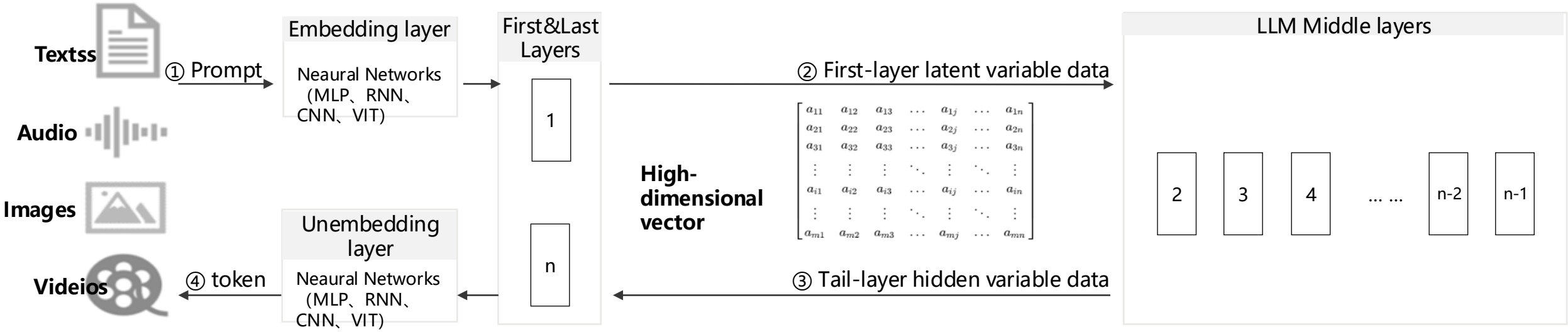
Model deployment plan: Deploy the Embedding & Unembedding layers and the model's first and last layers at the edge, and deploy the model's middle layers in the cloud intelligent computing center.



Data does not leave the park
Prompt is entered locally, Token is generated locally

Transmission data is available but not visible
Intermediate transmission of hidden variable data, high-dimensional vectors changed by the model

Cloud Model Security
Intermediate layer deployed in the cloud, unable to restore the complete model



Thank you.